23 Nitrous Oxide Plant Example (content warning: #explosion)

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9:13 PM

Reaction: $NH_4NO_3\,(l) \rightarrow N_2O\,(g) + 2H_2O(g)$

$A\,(l) \rightarrow B\,(g) + 2W(g)$

CSTR:

$\dot{Q} + \dot{W}_s = 0?$

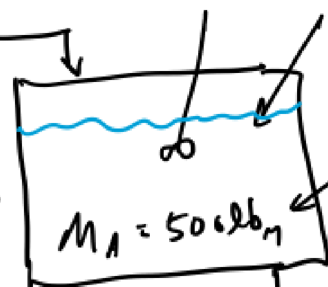
$\dot{m}_{A0}$ 83 wt%

$\dot{m}_{W0}$ 17 wt%

liquid

$T_0 = 200°F$

$\dot{m}_{T0} = 310\ \frac{lbm}{hr}$



$X = 0.9999$

↑ ?

$\dot{m}_A$ (g)

$\dot{m}_W$ (g)

$\dot{m}_B$ (l)

$T = 510°F$

$-r_A = kM_A$, units $= \frac{lb_m}{min}$

mass balance:

$$0 = \dot{m}_{A0} - \dot{m}_A - kM_A$$

$$\therefore M_A = \frac{\dot{m}_{A0} - \dot{m}_A}{k}$$

$k = 0.00517\ min^{-1}$ @ $510°F = 0.307\ hr^{-1}$

$$\dot{m}_{A0} = \left(310 \frac{lbm}{hr}\right)(0.83)$$

$$\dot{m}_A = \dot{m}_{A0}(1 - X) = \left(310 \frac{lbm}{hr}\right)(0.83)(1 - 0.9994)$$

$$\therefore M_A = \frac{\left(310 \frac{lbm}{hr}\right)(0.83)(1 - (1 - 0.9994))}{0.307\ hr^{-1}}$$

$M_A = 829\ lb_m$ A in the reactor

Analyze ΔH_{rxn}

$\Delta H_{rxn} = -336 \frac{Btu}{lbm}$ @ $500°F$ constant?

$C_{pA} = 0.38 \frac{Btu}{lb_m °F} = 30.4 \frac{Btu}{lbmol °F}$

$C_{pW} = 0.47 \frac{Btu}{lbm °F} = 8.5 \frac{Btu}{lbmol °F}$

$C_{pB} = ?$ $B = N_2O$. Assume it's

an ideal gas. For a polyatomic ideal gas, $C_p = 4R = 4\left(1.986 \frac{Btu}{lbmol \cdot °F}\right) = 7.9 \frac{Btu}{lbmol \cdot °F} = 0.18 \frac{Btu}{lb_m \cdot °F}$

$\Delta C_p = C_{pB} + 2C_{pW} - C_{pA} = -5.5 \frac{Btu}{lbmol \cdot °F} = -0.07 \frac{Btu}{lb_m A \cdot °F}$ ← very small compared to ΔH_{rxn}, so assume ΔH_{rxn} is constant w/ T

energy balance:

$$0 = \dot{m}_{A0} H_{A0} + \dot{m}_{W0} H_{W0} - \dot{m}_A H_A - \dot{m}_W H_W - \dot{m}_B H_B + (\dot{Q} + \dot{W}_s)$$

$$\dot{m}_W = \dot{m}_{W0} + \frac{2\dot{m}_{A0}}{MW_A} MW_W = \dot{m}_{A0}\left(\Theta_W + 2\frac{MW_W}{MW_A} X\right), \quad \Theta_W = \frac{\dot{m}_{W0}}{\dot{m}_{A0}} = \frac{0.17}{0.83}$$

$$0 = \dot{m}_{A0} H_{A0} + \dot{m}_{W0} H_{W0} - \dot{m}_{A0}(1-X) H_A$$

$$- \dot{m}_{A0}\left(\Theta_W + 2\frac{MW_W}{MW_A} X\right) H_W - \dot{m}_{A0} H_{N2O}\left(\frac{MW_{N2O}}{MW_A}\right) X$$

$$+ (\dot{Q} + \dot{W}_s)$$

$$0 = \dot{m}_{A0}(H_{A0} - H_A) + \dot{m}_{W0}(H_{W0} - H_W)$$

$$+ \dot{m}_{A0} X\left(H_A - 2\frac{MW_W}{MW_A} H_W - \frac{MW_{N2O}}{MW_A} H_{N2O}\right) + (\dot{Q} + \dot{W}_s)$$

$$0 = \dot{m}_{A0} C_{pA}(T_0 - T) + \dot{m}_{W0}(H_{W0} - H_W) +$$

$$\dot{m}_{A0} X(-\Delta H_{rxn}) + (\dot{Q} + \dot{W}_s)$$

get $H_{w0} - H_w$:

feed :

$T = 200°F$

liquid

exit :

$T = 510°F$

gas

$$\underbrace{\begin{matrix}\text{liquid}\\ 200°F\end{matrix} \longrightarrow \begin{matrix}\text{liquid}\\ 212°F\end{matrix}}_{\text{neglect}} \longrightarrow \begin{matrix}\text{gas}\\ 212°F\end{matrix} \longrightarrow \begin{matrix}\text{gas}\\ 510°F\end{matrix}$$

$$\therefore \; H_{w0} - H_w = H^{\circ}_{f,w(l)} - H^{\circ}_{f,w(g)} + C_{pw}(212°F - 510°F)$$

$$\therefore \; 0 = \dot{m}_{A0} C_{pA}(T_0 - T) + \dot{m}_{w0}\left(H^{\circ}_{f,w(l)} - H^{\circ}_{f,w(g)}\right) + \dot{m}_{w0} C_{pw}(212°F - 510°F) + \dot{m}_{A0} X(-\Delta H_{rxn}) + (\dot{Q} + \dot{W}_s)$$

$\therefore \quad \dot{Q} + \dot{W}_s = 5739 \ \frac{Btu}{hr} = 2.25 \ hp$

<u>feed shut off:</u>

model like an adiabatic batch reactor:

$T_0 = 510\,°F$

<u>mass balance:</u> $M_{A0} \frac{dX}{dt} = k M_A = k M_{A0} (1 - X)$

$$\therefore \quad \frac{dX}{dt} = k(1-X)$$

$$k = (0.00517 \ min^{-1}) \exp\left(44520\,°R \left(\frac{1}{510 + 460\,°R} - \frac{1}{T + 460\,°R}\right)\right)$$

<u>energy balance:</u>

reference T for ΔH_{rxn}

$$T = \frac{T_0 \sum \theta_i C_{pi} + X(-\Delta H_{rxn} + \Delta C_p \cdot 500\,°F)}{\sum \theta_i C_{pi} + X \Delta C_p}$$

$$\sum \theta_i C_{pi} = 0.38 \frac{Btu}{lb_m A\,°F} + \frac{0.17 \ lb_m W}{0.83 \ lb_m A}\left(0.47 \frac{Btu}{lb_m W\,°F}\right)$$

$$= 0.48 \frac{Btu}{lb_m A\,°F}$$